MIL-B-21442B(OS)
AMENDMENT 5
26 Oct 1970

Superseding
AMENDMENT 4
24 March 1966

MILITARY SPECIFICATION

BATTERY, STORAGE, NICKEL-CADMIUM

This amendment forms a part of Military Specification MIL-B-21442B(OS) of 13 August 1963 and has been approved by the Naval Ordnance Systems Command, Department of the Navy

By this change basic specification MIL-B-21442B(Wep) is changed to read MIL-B-21442B(OS).

Wherever it appears in specification delete "Bureau of Naval Weapons" and substitute "Naval Ordnance Systems Command"; also delete "BUWEPS" and substitute "NAVORD".

Page 5, paragraph 3.5.3: Delete "and product characteristics".

Page 6. Add the following new paragraph:
"3.5.8 Charge. There shall be no spewing of electrolyte during charge."

Page 6, paragraph 4.1. Delete and substitute:
"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure that supplies and services conform to prescribed requirements."

Page 8, paragraph 4.4.4.3: Delete and substitute:
"4.4.4.3 Charging. After its preparation (4.4.4.2), charge the battery using the constant voltage method (3.4.2.2) and battery charger (4.3.1). The vent valves shall allow the escape of gas without the loss of electrolyte during the charging period."

Page 9, paragraph 4.4.4.5. Delete and substitute:
"4.4.4.5 Cycling test. Repeat both the charging test (4.4.4.3) and discharge test (4.4.4.4) twice, except that the first time the charging test is repeated the battery shall be charged by the constant current

FSC 6140

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:36Z
Check the source to verify that this is the current version before use.

method (3.4.2.2). The second time the charging test is repeated the battery shall be charged by the constant voltage method (3.4.2.2)."

Page 9, paragraph 4.5.4, line 3: Delete "and product characteristics".

Page 10, paragraph 4.5.4.1. Delete and substitute:
"4.5.4.1 Low temperature operation. The battery shall be charged using the constant current voltage method (3.4.2.2) and the battery charger (4.3.1) and placed in a test chamber at 0 degrees F plus 5 minus 0 degrees F for a period of 48 plus or minus 5 hours. At the end of this period test the battery as follows:"

Page 10, paragraph 4.5.4.2: In line 1 delete "The battery shall be placed..." and substitute "The battery shall be charged using the constant current voltage method (3.4.2.2) and the battery charger (4.3.1) and shall be placed..."; in lines 6 and 7 delete "and product characteristics".

Page 10. Delete Table I and substitute:

Table I

Cycle	Type	Requirements
1	Measured	3.4.1
2-10	Automatic	None
11	Constant Current	None
12-19	Automatic	None
20	Measured	3.4.1
21-29	Automatic	None
30	Constant Current	None
31-39	Automatic	None
40	Measured	3.4.1
41-49	Automatic	None
50	Constant Current	None
51-59	Automatic	None
60	Measured	3.4.1
61-69	Automatic	None
70	Constant Current	None
71-79	Automatic	None
80	Measured	3.4.1
81-89	Automatic	None
90	Constant Current	None
91-99	Automatic	None
100	Measured	3.4.1, except that seven plus one minus zero shall be acceptable

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:36Z
Check the source to verify that this is the current version before use.

Page 11. Delete paragraph 4.5.5.3 and substitute:

"4.5.5.3 A constant current cycle shall consist of a constant current charge (3.4.2.2) followed by a constant resistance discharge (using 4.3.4) for eight plus one minus zero minutes. The discharge cycle shall be followed by a rest period as described in 4.4.4.2.

4.5.5.4 If the battery fails to meet the capacity requirements of a measured cycle (Table I), as many as four repeat tests shall be permitted to demonstrate conformance to capacity requirements as specified in 3.4.1. Repeat tests shall not be counted as life cycles."

Page 12: Delete "NOTICE" paragraph.

Preparing Activity
Navy - OS
Project No. 6140-N 290

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T09:36Z
Check the source to verify that this is the current version before use.